Minghe Zhang

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Education

Georgia Institute of Technology — Atlanta, Georgia, USA
Ph.D. in Machine Learning — Aug. 2019 – May 2023
- Department: H. Milton Stewart School of Industrial and Systems Engineering (ISYE)
- Advisor: Dr. Yao Xie

Georgia Institute of Technology — Atlanta, Georgia, USA
M.S. in Electrical and Computer Engineering — Aug. 2017 – May 2019
- Thesis: "Online Subspace Tracking for Community Change Detection"

Tsinghua University — Beijing, China
B.S. in Electronic Engineering — Aug. 2013 – May 2017
- Thesis: "Hardware Acceleration on Image Recognition"

Skills

- **Programming:** Python, PyTorch, TensorFlow, SQL, MATLAB, R, C/C++, Java, JavaScript, HTML, CSS.
- **Technical:** Large Language Model (LLM) Evaluation, Recursive Self-Improvement (RSI), Agentic Self-Learning, Deep Learning Ranking & Recommendation, Meta-Learning, Time-Series Analysis, Point Process, Anomaly Detection.
- **Mathematics:** Stochastic Processes, Statistical Machine Learning, Sequential Change Detection, Causal Inference, Convex Optimization.

Work Experience

Google — Kirkland, WA, USA
Machine Learning Engineer — 2026 – Present
- Focusing on Recursive Self-Improvement (RSI) and Large Language Model (LLM) Evaluation.
- Designing multi-round agentic self-learning pipelines, statistical evaluation harnesses, and high-reliability LLM-as-a-Judge methodologies for enterprise AI agents.

Ads Ranking & Recommendation, Meta — Bellevue, WA, USA
Research Scientist — May 2024 – 2026
- Researched and scaled large-scale deep learning ranking and recommendation algorithms for Meta Ads production systems.
- Led end-to-end model architecture iteration, online A/B experimentation, and metric governance to improve ad relevance and conversion efficiency.

E-Commerce Recommendation Team, TikTok — Bellevue, WA, USA
Machine Learning Scientist — July 2023 – May 2024
- Worked on large-scale short-video and e-commerce recommendation systems.

International Machine Learning, Amazon — Seattle, WA, USA
Applied Scientist Intern — Fall 2022
- Accelerating training DNN through data subsampling.
- Proposed an adaptive subsampling method called *Adaselection* for deep learning models in large-scale machine learning systems.
- Demonstrated the method to be effective on various classification, regression, and natural language processing tasks when compared with industry-wide adopted baselines.

Recommendation Core ML, Meta Menlo Park, CA, USA
Machine Learning Engineering Intern Summer 2022

- Improving Early Stage Ranking Model in Instagram Reels.
- Explored improving the early-stage ranker from multiple perspectives, including building overarch on the TTSN model and using Knowledge Distillation to transfer knowledge from the final ranker.
- Achieved improved NE metric on the early-stage ranker for the IG Reels model.

Tencent AI Lab, Machine Learning Center Shenzhen, China
Research Assistant supervised by Dr. Long-Kai Huang Summer 2021

- Out-of-Distribution Detection for Task-Adaptive Meta Learning.
- Based on MAML, developed an out-of-distribution task by combining data from various distributions.
- Proposed a multi-stage optimization process with learning-to-stop criteria to improve fine-tuning.

Argonne National Lab Chicago, IL, USA
Research Assistant supervised by Dr. Qiu Feng Fall 2020

- Graph Learning for solving Combinatorial Optimizations.
- Worked on using a graph embedding technique to optimize a power restoration problem for better generalization ability.

SF Express Shenzhen, Guangzhou, China
Machine Learning and Artificial Intelligence Engineer Summer 2018

- Built a recommendation system based on logistic networks between companies.
- Implemented change point detection to analyze fluctuations in employee salaries.
- Applied SVM, random forest, and deep neural networks to predict value-added delivery insurance demand.

Research Experience

Georgia Institute of Technology Atlanta, GA, USA
Research Assistant supervised by Dr. Yao Xie Spring 2020 – Spring 2023

- Spatial-Temporal Mutually Interactive Process Models for Solar Radiation Anomaly Events.
- Built a generative spatial-temporal Bernoulli model to fit extracted abnormal events in solar radiation observations.
- Developed an online learning algorithm to calibrate thresholds for predicting future abnormal events, achieving significantly higher accuracy under non-stationary settings.
- Conference version presented at INFORMS Conference on Service Science (ICSS 2021).

Georgia Institute of Technology Atlanta, GA, USA
Research Assistant supervised by Dr. Yao Xie Fall 2018 – Spring 2023

- Spectral CUSUM for Online Network Structure Change Detection.
- Designed a Spectral CUSUM procedure to detect community change in dynamic graphs.
- Used a Gaussian model to represent graph structure and derived closed-form expressions for Expected Detection Delay (EDD) and Average Run Length (ARL).
- Published conference paper at ICASSP 2020 and journal version in *IEEE Transactions on Information Theory*.

Georgia Institute of Technology Atlanta, GA, USA
Research Assistant supervised by Dr. Yao Xie Spring 2020

- Deep Attention Point Processes with Neural Spectrum Fourier Kernel.
- Created a novel attention-based model for discrete event data to capture complex non-linear temporal dependence structures.
- Incorporated an attention mechanism into the conditional intensity function of point processes with a neural spectrum Fourier kernel embedding.
- Published at AISTATS 2021 (oral presentation, acceptance rate: 3.1%).

Publications

[1] **M. Zhang**, L. Xie, and Y. Xie, "Spectral-CUSUM for network structure change detection," *IEEE Transactions on Information Theory*, 2023.

[2] S. Zhu, S. Yuchi, **M. Zhang**, and Y. Xie, "Sequential adversarial anomaly detection for dependent events," *INFORMS Journal on Data Science (IJDS)*, 2023.

[3] D. Guo, L. Tian, **M. Zhang**, M. Zhou, and H. Zha, "Learning prototype-oriented set representations for meta-learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[4] S. Zhu, L. Xie, **M. Zhang**, R. Gao, and Y. Xie, "Distributionally robust weighted k-nearest neighbors," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[5] S. Wei, S. Zhu, **M. Zhang**, and Y. Xie, "Goodness-of-fit test for mismatched self-exciting processes," in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 2021, pp. 1243–1251.

[6] **M. Zhang**, C. Xu, and Y. Xie, "Spatial-temporal mutually interactive process models for solar radiation anomaly events," *The 2021 INFORMS Conference on Service Science (ICSS 2021)*, 2021.

[7] S. Zhu, H. S. Yuchi, **M. Zhang**, and Y. Xie, "Sequential adversarial anomaly detection with deep Fourier kernel," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2021, pp. 3345–3349.

[8] S. Zhu, **M. Zhang**, R. Ding, and Y. Xie, "Deep Fourier kernel for self-attentive point processes," in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 2021, pp. 856–864.

[9] **M. Zhang**, L. Xie, and Y. Xie, "Online community detection by spectral CUSUM," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2020, pp. 3402–3406.

[10] S. Zhu, R. Ding, **M. Zhang**, P. Van Hentenryck, and Y. Xie, "Spatio-temporal point processes with attention for traffic congestion event modeling," *IEEE Transactions on Intelligent Transportation Systems*, 2020.

[11] H. Cheng, H.-L. Wang, **M. Zhang**, D. Hand, and P. A. Beerel, "Automatic retiming of two-phase latch-based resilient circuits," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 38, no. 7, pp. 1305–1316, 2018.

[12] H.-L. Wang, **M. Zhang**, and P. A. Beerel, "Retiming of two-phase latch-based resilient circuits," in *Proceedings of the 54th Annual Design Automation Conference (DAC)*, 2017, pp. 1–6.